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Diff4MMLiTS: Advanced Multimodal Liver Tumor Segmentation via Diffusion-Based Image Synthesis and Alignment

Shiyun Chen[1†], Li Lin[1,2†], Pujin Cheng[1,2], Zhicheng Jin[3], Jianjian Chen[3], Haidong Zhu[3], Kenneth K. Y. Wong[2], and Xiaoying Tang[1,4,*]

[1] Department of Electronic and Electrical Engineering, Southern University of Science and Technology, Shenzhen, China
tangxy@sustech.edu.cn
[2] Department of Electrical and Electronic Engineering, The University of Hong Kong, Hong Kong SAR, China
[3] Department of Radiology, Zhongda Hospital, Medical School, Southeast University, Nanjing, China
[4] Jiaxing Research Institute, Southern University of Science and Technology, Jiaxing, China

Abstract. Multimodal learning has been demonstrated to enhance performance across various clinical tasks, owing to the diverse perspectives offered by different modalities of data. However, existing multimodal segmentation methods rely on well-registered multimodal data, which is unrealistic for real-world clinical images, particularly for indistinct and diffuse regions such as liver tumors. In this paper, we introduce Diff4MMLiTS, a four-stage multimodal liver tumor segmentation pipeline: pre-registration of the target organs in multimodal CTs; dilation of the annotated modality's mask and followed by its use in inpainting to obtain multimodal normal CTs without tumors; synthesis of strictly aligned multimodal CTs with tumors using the latent diffusion model based on multimodal CT features and randomly generated tumor masks; and finally, training the segmentation model, thus eliminating the need for strictly aligned multimodal data. Extensive experiments on public and internal datasets demonstrate the superiority of Diff4MMLiTS over other state-of-the-art multimodal segmentation methods.

Keywords: Multimodal Learning · Liver Tumor Segmentation · Diffusion-Based Synthesis.

1 Introduction

Liver tumor segmentation is crucial for clinical diagnosis and treatment. Previous studies have achieved notable results in unimodal liver tumor segmentation, confirming the effectiveness of deep learning techniques in this area [5,9]. As different

† These authors contributed equally.

types of tumors exhibit distinct characteristics across various modalities, clinical radiologists often use multimodal CTs for more comprehensive diagnosis and more precise delineation of tumor boundaries [28]. Integrating multimodal information significantly enhances segmentation performance [13,26,27]. However, compared to unimodal, the high annotation costs and data scarcity significantly constrain the exploration of multimodal segmentation methods.

Recent multimodal learning studies have underscored the importance of data alignment [4,12]. Alignment is equally vital in multimodal medical image segmentation tasks. Public multimodal benchmarks like BraTS [14] and AutoPET [6] either employ standardized anatomical templates for rigid alignment or are collected simultaneously, inherently ensuring rigid data alignment. Thus, most current multimodal tumor segmentation methods operate under the assumption of strict data alignment [22, 25]. In contrast, in real clinical settings, multimodal CTs often exhibit significant misalignment due to spatiotemporal differences in data acquisition. Although some registration methods based on traditional paradigms [10] or deep learning [2, 11] have been proposed and achieved reasonable registration at the organ or tissue level, they still struggle to fully align the diffuse and indistinct tumor regions. This presents unavoidable noise for existing multimodal segmentation algorithms.

In recent years, due to the improvement of generative models, tumor segmentation based on synthesis has become a highly regarded research topic [19, 24]. Previous studies have demonstrated the effectiveness of using synthetic images to enhance liver tumor segmentation [3,8,23]. However, these methods are primarily limited to unimodal images and fail to achieve expected performance when directly applied to multimodal tasks. Nevertheless, these efforts provide two key insights: first, synthetic tumor samples can be used to alleviate the challenge of data scarcity and enhance model generalization; second, synthesizing rigorously aligned multimodal CTs can partially mitigate the current dependence of segmentation algorithms on well-registered multimodal data.

In this context, we present an innovative and effective multimodal segmentation framework named Diff4MMLiTS. Specifically, the process begins with preliminary registration and data preprocessing, during which the target organ is roughly aligned, but the tumor remains misaligned. Therefore, we employ the mask of the annotated modality, dilating it to ensure adequate coverage of the tumor across various modalities. Subsequently, this mask is utilized in the Normal CT Generator (NCG) module for inpainting, yielding the normal CT. Within the latent space, the Multimodal CT Synthesizer (MCS) module leverages the multimodal CT features and randomly generated tumor masks to synthesize strictly aligned multimodal CTs with tumors. The aforementioned synthesis process employs the common mask, yet each modality's CT is synthesized according to its unique feature distribution mapping. Finally, the roughly aligned real data is combined with the fully aligned synthetic data to train the segmentation model. Our contributions are as follows:

- An innovative multimodal liver tumor segmentation framework, Diff4MMLiTS, is proposed to effectively leverage unaligned multimodal liver tumor images.

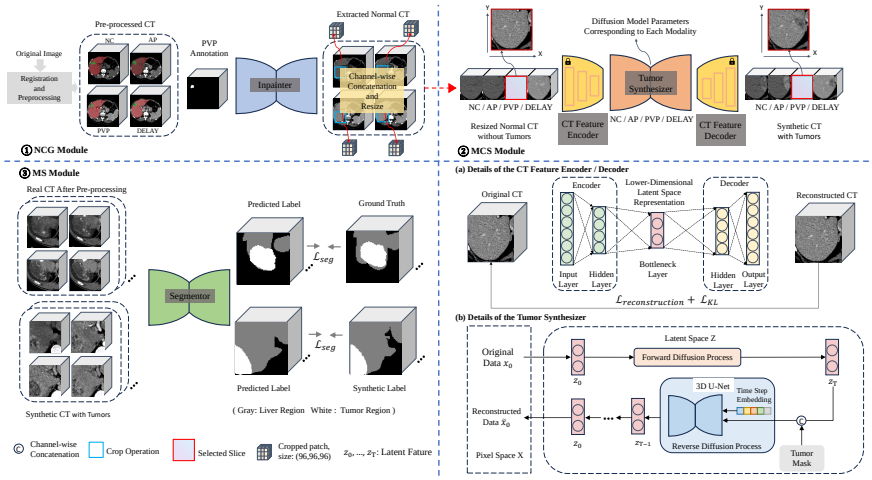

Fig. 1. The architecture of Diff4MMLiTS. The NCG module uses the extended PVP mask to inpaint multimodal images to acquire normal CTs. The MCS module uses normal CTs to synthesize multimodal CTs. The MS module trains segmenter using real and synthetic data. MCS comprises two components: a 3D autoencoder consisting of a CT feature encoder and decoder, a tumor synthesizer based on a diffusion model.

- The designed Inpainting-Synthesis-Segmentation pipeline leverages a diffusion model to synthesize a large number of perfectly aligned multimodal liver tumor images, thereby significantly improving segmentation performance.
- Diff4MMLiTS is evaluated on both internal and external datasets, with experimental results demonstrating outstanding performance.

2 Method

Diff4MMLiTS comprises three pivotal components: the NCG module, the MCS module, and the MS module. These components are trained sequentially, with each preceding stage's output serving as input for the next. NCG provides the CT background for MCS to synthesize CTs with tumor. MCS then supplies MS with synthetic images to train the segmenter. The following subsections will provide a detailed elaboration. The overall framework is shown in Fig. 1.

2.1 Normal CT Generator (NCG) Module

Since there is no publicly available paired multimodal normal CT dataset, NCG is used to generate multimodal normal CT as the basis for tumor synthesis. Inspired by Suvorov [21], we employ an inpainting network based on Fast Fourier Convolution (FFC). To reduce computational load, we perform slice-by-slice inpainting of 2D slices within 3D CT scans. Given an image x and a binary mask of the inpainting regions m, they are stacked as follows:

$$x' = \text{stack}(x \odot (1 - m), m), \tag{1}$$

where $x \odot (1-m)$ denotes the masked image. The feed-forward inpainting network $f_\theta(\cdot)$ processes the input x' in a fully convolutional manner and generates the inpainted image $\hat{x} = f_\theta(x')$. Specifically, the image and mask pair serve as network inputs, which are downsampled into FFC residual blocks. During FFC processing, the input is divided into two parallel branches: a local branch, using traditional convolution, captures local information, while a global branch, based on real FFT, acquires global contextual information. These two outputs are then cross-fused and channel-wise concatenated to produce the final output.

We fine-tune the model using pre-trained parameters and followed the training strategy of Suvorov et al. [21]. For the fine-tuning of NCG, we employ original internal multimodal data to adapt it to the inpainting of multimodal CTs. After training, the inpainter is utilized to remove tumor foreground regions from multimodal CT with tumors. For each tumor region in the portal venous phase (PVP) mask, a 5×5 kernel is first employed for morphological closing, followed by a single iteration of dilation using a 3×3 kernel, to mitigate subjective noise and ensure maximal coverage of tumor regions across all modalities. NCG used the dilated PVP noise annotation as the mask to remove the foreground from labeled samples, ultimately producing a corresponding normal CT for each CT.

2.2 Multimodal CT Synthesizer (MCS) Module

MCS is divided into two stages: first, a autoencoder is trained to learn comprehensive latent features, including the CT feature encoder $\mathcal{E}$ and decoder $\mathcal{D}$; second, a tumor synthesizer based on a latent diffusion model (LDM) [20] is trained to generate synthetic images. This latent feature-driven diffusion process enables efficient and reliable tumor generation within a compact latent space.

Specifically, the VQGAN-based autoencoder performs the encoding and reconstruction between the image x_0 and its latent representation z_0, giving $\tilde{x}_0 = \mathcal{D}(z_0) = \mathcal{D}(\mathcal{E}(x_0))$. Since the universal autoencoding stage only requires training once and can be utilized repeatedly for multiple diffusion model trainings [20], we adapt the pre-trained autoencoder [3]. The diffusion process comprises a forward process and a reverse process. The forward diffusion process transforms the data distribution into a standard Gaussian distribution at each time step $t \in \left[1, \cdots, T\right]$ by the following process:

$$q\left(\boldsymbol{z}_t \mid \boldsymbol{z}_0\right) = \mathcal{N}\left(\boldsymbol{z}_t \mid \sqrt{\bar{\alpha}_t}\boldsymbol{z}_0, \left(1-\bar{\alpha}_t\right)\boldsymbol{\epsilon}\right), \tag{2}$$

where ϵ is the random noise, $\bar{\alpha}_t$ denotes the noise level.

The reverse generation process aims to gradually generate data samples from the noise by the following process:

$$p\left(z_0\right) = p\left(z_T\right) \prod_{t=1}^{T} p\left(z_{t-1} \mid z_t, y\right), \tag{3}$$

where y is the tumor mask for conditioning information. During the inference stage, a coordinate within the liver region is randomly selected as the center

point, and random values for the semi-axes x, y, and z are assigned. An ellipsoidal mask approximating the tumor is then generated based on these parameters, followed by elastic deformation to produce the tumor mask y, indicating the location of the synthesized tumor. During the training stage, $\epsilon_\theta(\cdot)$ denotes the denoising encoder, the training objective of LDM can be written as:

$$\mathcal{L}_{LDM} := \mathbb{E}_{\mathcal{E}(x_0),\epsilon\sim\mathcal{N}(0,1),t}\left[\|\epsilon - \epsilon_\theta(z_t, y, t)\|_2^2\right]. \tag{4}$$

2.3 Multimodal Segmentation (MS) Module

The segmenter is a U-shape network. During training, MS dynamically synthesizes CTs with tumors from inpainted CTs, thereby constructing a hybrid dataset that combines synthetic and real data. The training objective of the segmenter is to minimize the voxel classification error relative to the ground truth. The loss function $\mathcal{L}_{seg}$ is consist of the Dice loss $\mathcal{L}_{dice}$ and the cross-entropy loss $\mathcal{L}_{ce}$:

$$\mathcal{L}_{seg} = \mathcal{L}_{dice} + \gamma\mathcal{L}_{ce}, \tag{5}$$

where $\gamma = 0.5$ is a factor for balancing the two loss contributions. During inference, if the input does not consist of all four modalities, certain modalities will be duplicated to match the input size required by the model.

3 Experiments and Results

3.1 Datasets

To the best of our knowledge, there is currently no publicly available well-aligned multimodal liver tumor CT dataset. Therefore, we construct an in-house dataset from the Second Affiliated Hospital of Southeast University, called mmLiTS. mmLiTS includes 45 sets of four-phase abdominal CTs with in-plane resolution of 512×512, spacing from 0.6445 mm to 0.9160 mm, slice counts ranging from 45 to 561, and thicknesses between 1.0 mm and 5.0 mm. Liver tumors are manually annotated by experienced radiologists, and the dataset is randomly split into training, validation, and test sets in a 3:1:1 ratio. Given that radiologists often rely on PVP images to identify tumor margins that are less discernible in other phases, annotations are provided for PVP. To further assess generalizability, we evaluate our method on the LiTS dataset [1]. As LiTS is a unimodal dataset without alignment issues, we replicate each image four times to match the input format required by our segmentation model.

3.2 Implementation Details

Diff4MMLiTS is implemented by PyTorch on a workstation equipped with NVIDIA RTX 3090. We use nnUNet as the default backbone. To avoid over-fitting, online data augmentations, including random flipping, rotation, and color jittering, are applied. All methods are trained for 1000 epochs. The preprocessing

Table 1. Quantitative comparisons of Diff4MMLiTS with state-of-the-art methods on mmLiTS. The best results are highlighted in bold.

Methods	DSC(%)	JAC(%)	SE(%)	PRE(%)
16'U-Net [5]	75.46	61.29	68.12	87.69
16'V-Net [15]	75.97	62.70	**81.78**	74.40
18'AttentionUNet [17]	67.80	53.75	65.25	77.72
18'nnUNet [9]	76.34	62.77	69.84	86.47
22'SwinUNETR [7]	77.28	63.41	71.72	86.02
18'SegResNet [16]	77.73	64.05	75.29	81.87
21'MAML [27]	76.06	63.08	67.92	**91.27**
22'mmformer [26]	77.32	64.32	78.76	77.94
23'A2FSeg [22]	69.74	55.90	61.93	88.66
24'PA-Net [13]	76.56	63.09	74.38	82.23
Diff4MMLiTS	**79.02**	**66.33**	74.47	85.24

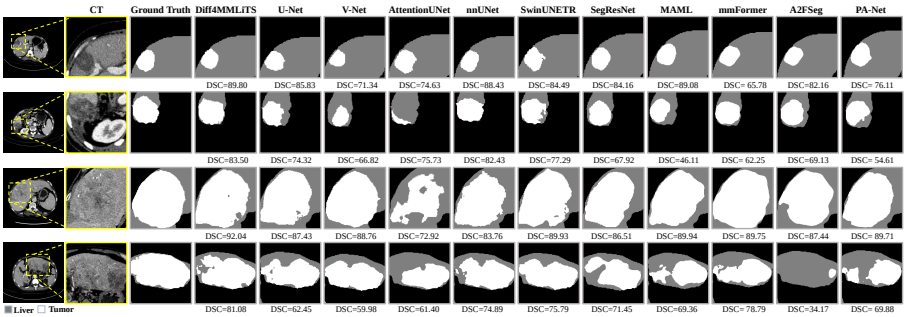

Fig. 2. Qualitative visualization results.

involves clipping the images to a [-21, 189] window and normalizing them to zero mean and unit standard deviation, followed by random cropping of 96×96×96 patches as input. The postprocessing filters the detection results by retaining only tumors located within the liver region.

3.3 Overall Performance

To evaluate our method, we compare Diff4MMLiTS with various general and multimodal segmentation methods. Given that achieving precise liver segmentation is relatively straightforward, we focus specifically on the segmentation of intrahepatic tumors. The comparison results are presented in Table 1, with visualizations shown in Fig. 2.

Compared to the U-Net and its variants, our method enhances the DSC by 1.74% over the second-best SwinUNETR. When compared with other multimodal medical image segmentation models, our method enhances DSC by 1.29% over the second best SegResNet. Moreover, Diff4MMLiTS significantly reduces false positives and false negatives, as reflected in the improved SE and PRE.

Table 2. Comparison of experimental results on LiTS. The best results are highlighted in bold.

Methods	DSC(%)	JAC(%)	SE(%)	PRE(%)
nnUNet (real tumors)	41.63	27.37	55.51	35.56
Diff4MMLiTS	**57.75**	**42.99**	**59.41**	**64.59**

Table 3. Results of Diff4MMLiTS equipped with different backbone models. The best results are highlighted in bold.

Methods	DSC(%)	JAC(%)	SE(%)	PRE(%)
U-Net [5]	75.46	61.29	68.12	87.69
U-Net w/ Diff4MMLiTS	77.41	64.25	72.22	85.17
AttentionUNet [17]	67.80	53.75	65.25	77.72
AttentionUNet w/ Diff4MMLiTS	74.38	60.40	67.37	84.58
nnUNet [9]	76.34	62.77	69.84	86.47
nnUNet w/ Diff4MMLiTS	**79.02**	**66.33**	**74.47**	**85.24**

This indicates enhanced tumor detection capability and a diminished likelihood of missed or incorrect detections. Overall, our proposed method has excellent superiority and achieves the best performance in the multimodal liver tumor segmentation task.

3.4 Generalizability and Adaptability of Diff4MMLiTS

We evaluate the proposed method on a publicly available external dataset to assess its out-of-distribution generalization capability, demonstrating that the model trained with Diff4MMLiTS performs effectively without retraining. All methods are trained on mmLiTS and tested on lesion samples from LiTS. The comparison results are shown in Table 2. Compared to nnUNet fully trained on real data, Diff4MMLiTS with the synthesis strategy achieves a 16.12% improvement in DSC. This implies that training models with such reliable synthetic images can effectively mitigate the risk of overfitting to in-distribution samples, thereby enhancing their ability to generalize to out-of-distribution samples.

To further assess the adaptability of Diff4MMLiTS, we integrate three different backbone models into the MS module, with the results summarized in Table 3. The results demonstrate that our framework adapts well to these backbones, yielding notable performance gains. Compared to models trained solely on real data, those using the hybrid training strategy achieve DSC improvements of 1.95%, 6.58%, and 2.68%, respectively. These findings highlight both the flexibility of our framework across different backbones and the effectiveness of the hybrid training strategy in enhancing segmentation performance.

3.5 Effectiveness of Multimodal Synthesis Strategy

As shown in Table 4, we evaluate the effectiveness of our synthesis strategy on both multimodal and unimodal segmentation methods, using nnUNet as the

Table 4. Results using only a portion of real diseased CT and corresponding normal CT training. Note that uni is short for unimodal and multi is short for multimodal.

Ratio(diseased/normal)		DSC(%)	JAC(%)	SE(%)	PRE(%)
uni	100% (29/29)	71.24	56.08	64.31	85.11
multi	10% (3/3)	73.87	59.58	76.40	73.13
	30% (9/9)	75.26	60.53	70.24	84.14
	50% (15/15)	76.97	63.45	70.25	87.43
	70% (20/20)	78.49	65.47	75.66	82.47
	100% (29/29)	79.02	66.33	74.47	85.24

Table 5. Results of Diff4MMLiTS equipped with different inpainting or synthesis methods. $ and # denotes the replacement of the inpainter and synthesizer, respectively. The best results are highlighted in bold.

Methods	DSC(%)	JAC(%)	SE(%)	PRE(%)
Median ($)	74.79	60.75	**79.43**	73.37
CUT (#)	77.64	63.92	74.78	84.62
Diff4MMLiTS	**79.02**	**66.33**	74.47	**85.24**

segmentation backbone in all settings. Our proposed method achieves superior performance using only a small subset of multimodal data. Notably, with just 10% of the data (real samples and inpainted multimodal CTs from three patients), our method outperforms the unimodal model trained on fully annotated data by 2.63% in DSC. Furthermore, compared to fully supervised multimodal models summarized in Table 1, our method achieves comparable performance using only 70% of the training data. These results demonstrate the efficiency and effectiveness of Diff4MMLiTS in enhancing liver tumor segmentation.

3.6 Contribution of Different Modules to the Framework

To comprehensively validate the importance of each module, we explore alternative designs. Specifically, we replace the inpainter in NCG with a median filter. For MCS, we replace the LDM in MCS with CUT [18], which constructs the source domain by concatenating normal CTs with randomly generated tumor masks and learns a mapping to the target domain of real tumor CTs. The impact of these alternatives on overall performance is shown in Table 5. The results indicate that the framework with the original inpainter and synthesizer achieves the best performance. In comparison, CTs generated by the median filter introduce noise and disrupt the original details, thereby affecting the quality of subsequent tumor synthesis. Additionally, the tumor images synthesized by CUT are overly smooth, making the complementary features between multimodal tumors less pronounced, which, in turn, affects the performance of the segmentation model. In contrast, the inpainter and synthesizer proposed in our original framework not only preserve background details, but also generate reliable multimodal liver tumors, playing a crucial role in the liver tumor segmentation task.

4 Conclusion

This paper presents a novel multimodal liver tumor segmentation framework, named Diff4MMLiTS. Diff4MMLiTS is a diffusion-based pipline comprising of a Normal CT Generator, a Multimodal CT Synthesizer and a Multimodal Segmenter. Extensive experiments demonstrate the superiority of our framework. A potential future direction is to employ an all-in-one diffusion model paradigm for multimodal tumor synthesis while exploring advanced network designs and multimodal fusion strategies to further enhance performance.

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